

Personalized AI assistants could either promote or hinder the emergence of collective intelligence

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Abstract

Advances in artificial intelligence (AI) have enabled the wide use of personalized assistance for cognitive tasks, which enhanced individual performance on various fields. However, the impact of AI assistance on collective intelligence remains poorly understood. AI assistance may undermine the emergence of collective intelligence by reducing the individual diversity or introducing structured biases into a population. Here, we develop a human-AI collective prediction model mediated by social learning based on incentive structure by extending the existing works on collective intelligence. Our model uncovers that emergence of collective intelligence is depend on how human and AI interact. When AI works as if it is omniscient and gives assistance regardless of users' interests, collective intelligence will be promoted. However, when AI works as a chat-bot and gives assistance only based on users' interests, collective intelligence could be hindered due to diversity loss and overconfidence on AI. We showed that, to promote collective intelligence, penalizing or incentivizing AI usage is not reliable. Instead, elaborating more diversity-aware incentives could be the key to promote and maintain the collective intelligence.

Introduction

Collective intelligence (CI) describes a phenomenon in which a group’s ability to solve problems, make decisions, or adapt to changing environments exceeds each member’s ability [1, 2, 3, 4, 5]. This phenomenon, also referred to as wisdom of crowds, has thus served as a central mechanism underlying the success of diverse organizations, ranging from animal groups [6, 7, 8] and human societies [2, 9] to machine-learning systems [10, 11]. Understanding the conditions that promote and maintain CI therefore remains a fundamental question across social sciences [12], evolutionary biology [13], and computer science [14]. Addressing this question has become particularly timely as artificial intelligence (AI) has begun to reshape how individuals acquire information, make decisions, and contribute to collective outcomes [15].

The recent emergence of generative AIs and large language models (LLMs) has changed the way individuals engage in various cognitive tasks, ranging from simple everyday tasks to more complex tasks, including scientific discovery [16] and creative writing [17]. In particular, while the use of generative AI models, such as ChatGPT, has been rapidly increasing [18], the potential long-term impact of personal AI assistants on human society, especially the CI, remains uncertain. Some argue that AI can promote CI by enhancing individual expertise [19], and opinion aggregation [20]. Conversely, empirical evidence indicate that the use of AIs can limit the diversity of opinions [17, 21, 22], a key requirement for the emergence of CI [23, 24, 25].

Our ability to predict the long-term impact of personal AI assistants on human CI is further hampered by the apparent discrepancy in the incentives that shape AI uses. On one hand, many industrial sectors are pushing towards increased AI usages, where the number of “tokens” used is often perceived as a proxy for productivity [26, 27]. On the other hand, many academic institutions appear to discourage, and even penalize, AI uses, reflecting concerns about plagiarism, misinformation, and limited independence [28, 29]. Such discrepancy raises a question about whether directly incentivizing or penalizing AI usage will benefit or harm human CI. These observations further highlight the importance of identifying a robust incentive structure that can promote CI in the presence of personal AI assistants. Such incentive structure should seek to retain the benefits of AIs and prevent potential risk, such as preserving the diversity of individual opinions.

To address these questions, we extend previous approaches for modeling CI, where each individual makes independent predictions for the outcome from partial observations of underlying causal factors [12, 13, 30]. In particular, we allow each individual to independently consult with personal AI assistants and revise their predictions accordingly. The resulting model allows us to investigate the robustness of human CIs under different assumptions about the AI model’s ability to make predictions as well as incentive structures that are associated with the prediction task. In doing so, we assess the impact of directly incentivizing or penalizing AI uses on CI and further suggest a balanced incentive structure that can maintain CI in the presence of AI.

66 Main

67 Model

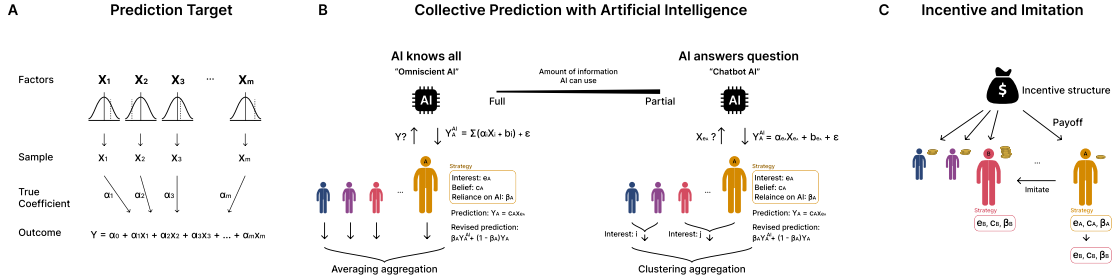


Figure 1: **A schematic diagram of the model for individual and collective prediction in the presence of personal AI assistants.** (A) The prediction target is modeled as a linear combination of causal factors. (B) Each player is characterized by three strategies: interest, belief, and reliance on AI. Players make predictions based on their interests and beliefs and subsequently revise them with AI assistance. We consider two types of AI, Chatbot AI and Omniscient AI, which differ in the amount of information available for generating predictions. (C) Players receive payoffs according to the incentive structure, and the population evolves through imitation of peers with higher payoffs.

68 We begin by extending previously established models of CI, in which a group of
 69 players participates in a prediction task (Figure 1): each player makes an individual
 70 prediction based on partial observations of the underlying causal factors, and the
 71 group forms a collective prediction by aggregating these individual predictions [30,
 72 12, 13]. Specifically, the prediction target (Y) is modeled as a linear combination of
 73 M causal factors x_i , with fixed coefficients α_i : $Y = \sum_{i=1}^M \alpha_i x_i + \alpha_0$, where each factor
 74 is drawn from independent normal distribution (Figure 1A). Then, each individual
 75 A observes a single factor of their interest x_{e_A} and makes a prediction based on
 76 their own belief c_A : $Y_A = c_A x_{e_A}$ (Figure 1B). We assume each individual can then
 77 independently consult their AI assistants and revise their prediction based on their
 78 reliance on AI β_A : $Y_A^{re} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A$ (Figure 1B). This revised prediction
 79 is aggregated to form collective prediction. **[SWP: I think we need to say averaging**
 80 **and clustering aggregation here?]**

81 To model AI, we assume that AI has already been trained on similar tasks and is
 82 able to make predictions based on partial or full observations of causal factors (Figure
 83 1B). Every time a player consults AI, it will make a new prediction Y_A^{AI} based on
 84 its internal parameters, which can include factor-specific bias (b_i) and stochastic
 85 error ϵ . Here, we consider two extreme types of AI response, which differ in the
 86 amount of information AI uses when making predictions. In one extreme, the AI
 87 can observe all causal factors and use all information to make predictions, regardless

88 of the user’s interest: $Y_A^{AI} = \sum_{i=0}^M (\alpha_i x_i + b_i) + \epsilon$. In the other extreme, AI works
 89 like a chatbot, where the AI predicts the outcome based only on the user’s interest:
 90 $Y_A^{AI} = \alpha_{e_A} x_{e_A} + b_{e_A} + \epsilon$. Hereafter, we refer to these two types as ‘Omniscient AI’
 91 and ‘Chatbot AI’, respectively.

92 Each generation consists of multiple round of predictions, and each player is re-
 93 warded a payoff according to the incentive structure (Figure 1C). For simplicity,
 94 we assume that a sufficiently large number of prediction rounds occurs within each
 95 generation, allowing us to use the expected payoff value rather than computing the
 96 payoffs for every round. We consider two incentive structures from previous work,
 97 “feedback” and “niche-expert”, which have been shown to promote near-perfect col-
 98 lective accuracy in the absence of AI [13]. The feedback structure gives higher payoff
 99 to individuals who can push the collective prediction toward the true target, regard-
 100 less of their individual predictive accuracy. [SWP: *maybe we need equations...*]
 101 In contrast, the niche-expert structure gives higher payoff to individuals who have
 102 both better predictive accuracy and a minority interest. [SWP: *maybe we need*
 103 *equations... but also say that this preserves diversity*] We note collective intelligence
 104 fails to emerge under the combination of averaging aggregation with niche-expert
 105 structure in the absence of AI [13].

106 Through time, the group evolves by allowing each individual to imitate peers
 107 who received higher payoffs (Figure 1C). Specifically, after each generation, two in-
 108 dividuals are randomly sampled, and one imitates all three strategies (the factor of
 109 interest, factor belief, and reliance on AI) of the other with a probability determined
 110 by the difference in their expected payoffs [SWP: *either refer to methods or supp.*].
 111 Individual predictions are aggregated into a collective prediction $\hat{Y}$ by simple averag-
 112 ing when using Omniscient AI and by clustering when using Chatbot AI. Collective
 113 accuracy is defined as the reduction in prediction error achieved by the collective
 114 prediction relative to a baseline prediction α_0 [SWP: *maybe need equation*]. Here-
 115 after, we refer to the near-perfect collective accuracy as accurate CI. [SWP: *How*
 116 *about: hereafter, we say that CI emerges when the collective accuracy reaches a near*
 117 *perfect level.*]

118 This model now allows us to explore the impact of AI on CI depending on the
 119 mode of AI and the incentive structure. Here, we begin by comparing the emergence
 120 of CI with and without AI and subsequently investigate a how different incentive
 121 structures affect the emergence CI in the presence of AI.

122 AI can either promote or prevent the emergence of CI

123 To understand the impact of AI, we first compare collective dynamics with and
 124 without AI. First, consistent with [13], simple imitation rules and incentive structures
 125 allow collective intelligence to emerge in the absence of AI (Fig. 2A,C,D). One
 126 exception is the use of simple averaging with niche-expert incentive structures, under
 127 which the collective accuracy of the population remains near zero [SWP: *because...*]
 128 (Fig. 2B) [SWP: *maybe need a supp figure*].

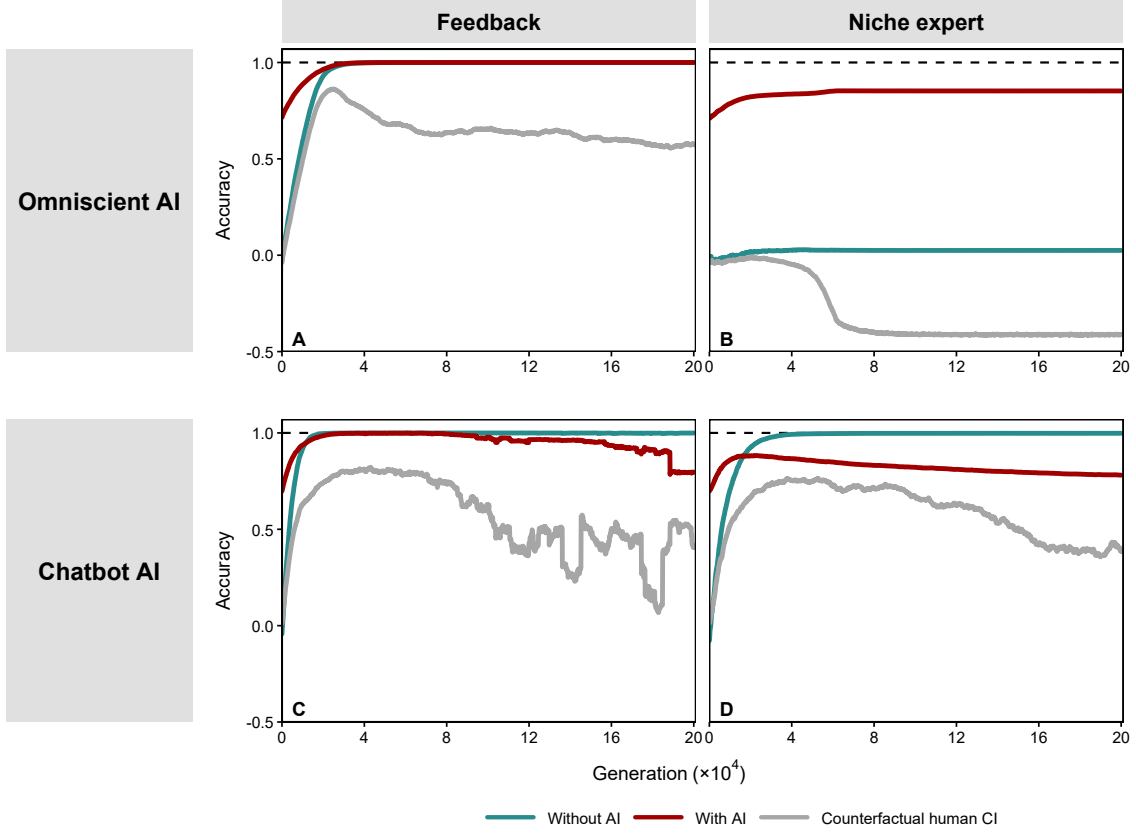


Figure 2: **Comparison of collective accuracy with and without AI.** Trajectories of collective accuracy when the group is assisted by the Omniscient AI (A,B) and Chatbot AI (C,D). [SWP: *Maybe we should also name the averaging rule below AI...*] Here, feedback (A,C) and niche-expert (B,D) incentive structures are considered. Green lines represent trajectories of collective accuracy without AI assistance. Red lines represent trajectories of collective accuracy with AI assistance. Gray lines represent trajectories of counterfactual collective accuracy, computed from players’ unrevised predictions, indicating the group’s predictive accuracy if AI assistance were unavailable at the given generation. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A, B), $c_A \sim \mathcal{N}(0, 5^2)$ for (C, D), and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.36$ and $\tau = 0.3$.

129 We find that AI usage can have heterogeneous outcomes depending on its type.
130 For example, the use of Omniscient AI can either preserve CI under the feedback
131 incentive structure (Fig. 2A) or promote higher collective accuracy under the niche-
132 expert incentive structure (Fig. 2B). These results are robust across different levels
133 of AI accuracy and can be shown analytically by assuming a sufficiently large group
134 of players (Fig. S2) [SWP: *probably need to swap S2 and S1 so that the current S2*

135 *appears first*].

136 In contrast, the use of Chatbot AI prevents the emergence of CI, thereby resulting
137 in lower levels of collective accuracy (Fig. 2C, D). Under the feedback incentive
138 structure, the use of Chatbot AI leads to a loss of diversity in the population’s
139 interest, making it impossible to account for effect of some causal factors [SWP:
140 *Supp*]. This causes an increase in collective variance and stochastic decrease in
141 collective accuracy [SWP: *Supp*]. We note that the loss of interest diversity may
142 arise even in the absence of AI in finite populations [SWP: *Supp*], but the use of
143 Chatbot AI makes the group more vulnerable to such loss [SWP: *Supp*]. Under
144 the niche-expert incentive structure, the population becomes overly reliant on AI,
145 causing the collective prediction to inherit the AI’s bias (Fig. S1). This leads to
146 an increase in collective bias and a decrease in collective accuracy [SWP: *Supp*].
147 Interestingly, we find that the niche-expert structure can prevent the use of AI under
148 specific assumptions about the relationship between the true coefficient and AI bias
149 (Fig. S2). In this case, the population completely excludes AI assistance, allowing
150 the CI to emerge eventually. However, this phenomenon can substantially delay the
151 emergence of CI [SWP: *Supp*].

152 To investigate whether the group can retain its predictive ability when AI assis-
153 tance suddenly becomes unavailable at any given time, we quantify counterfactual
154 collective accuracy based on unrevised predictions. Across all scenarios, we find that
155 the counterfactual collective accuracy is lower than the collective accuracy of the
156 corresponding population.

157 In summary, an omniscient AI can promote CI, but AI operating with limited in-
158 formation may prevent the emergence CI. Moreover, reliance on AI can limit inherent
159 ability of the population to make accurate predictions on their own, raising concerns
160 about future AI usage. These findings further underscore the need for incentive
161 structures that can mitigate the potential risks associated with AI assistance.

162 **Directly incentivizing or penalizing AI use can additionally** 163 **harm CI**

164 Given the strong influence of AI on collective outcomes, a straightforward interven-
165 tion is to directly incentivize or penalize the use of AI. [SWP: *maybe a sentence*
166 *about how this reflects real world scenarios.*] To investigate whether such direct
167 incentives or penalties can help the emergence of CI, we modify the two incentive
168 structures by adding a payoff term proportional to individual reliance on AI:

$$\text{Feedback/Niche-expert} + \lambda\beta_A \quad (1)$$

169 where λ represent the strength of intervention ($\lambda > 0$) or penalty ($\lambda < 0$). [SWP:
170 *maybe we need to call incentive pi earlier on...*] [SWP: *also good to define beta*
171 *again here as a reminder.*]

172 Directly incentivizing or penalizing AI use can further reduce the collective ac-
173 curacy (Fig. 3C, E, G), except when Omniscient AI is used under the feedback

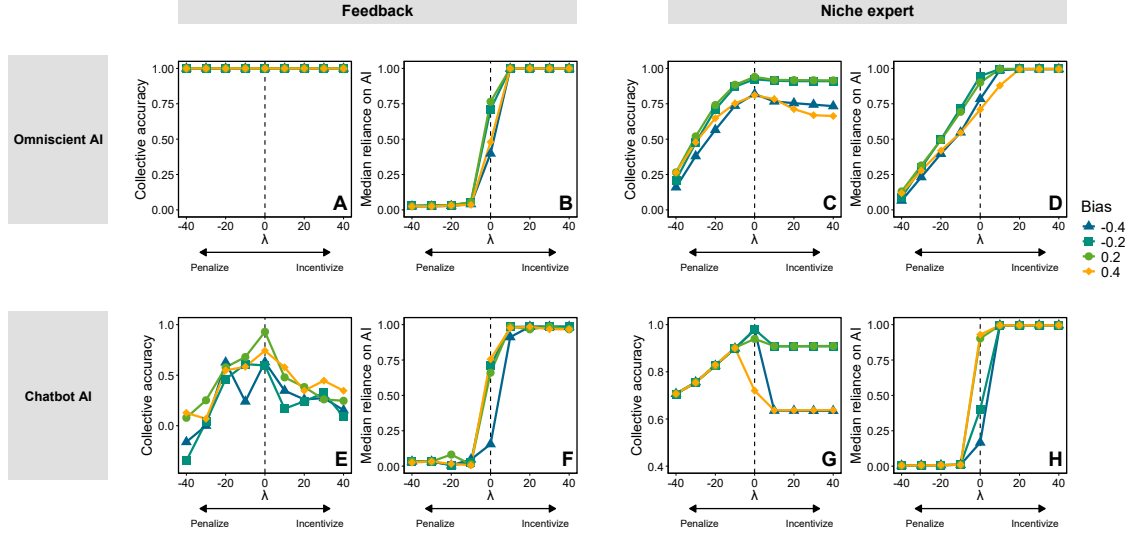


Figure 3: **Collective accuracy under the influence of direct intervention on AI use.** Collective accuracy and median reliance on AI under varying strengths of intervention (λ) across different assumptions about AI types (omniscient AI, A–D and chatbot AI, E–H) and incentive structures (Feedback, A,B,E,F, and Niche expert, C,D,G,H). Positive λ represents an incentive for AI use, whereas negative λ represents a penalty for AI use. In each panel, the vertical dashed line indicates the original incentive structure ($\lambda = 0$). Four AI bias levels are shown: $b_i = \pm 0.2$, representing higher-accuracy AI, and $b_i = \pm 0.4$, representing lower-accuracy AI. Collective accuracy (A,C,E,G) and median reliance on AI (B,D,F,H) are averaged over generations 190,000–200,000. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A–D), $c_A \sim \mathcal{N}(0, 5^2)$ for (E–H), and $\beta_A \sim U(0, 1)$. AI parameter: $\tau = 0.3$.

structure; in the latter case, the population retains collective intelligence regardless of the intervention strength (Fig. 3A). The reduction in collective accuracy arises because incentivizing or penalizing AI usage causes individuals to imitate others based on their AI usage rather than their predictive accuracy. Thus, stronger interventions can push the group to always ($\beta = 1$) or never ($\beta = 0$) rely on AI (Fig. 3B,D, F, H) but these scenarios do not necessarily lead to better outcomes (Fig. 3D, F, H). When AI bias is relatively large, a weak penalty can lead to higher collective accuracy, but this represents a narrow parameter regime (Figure 3G).

Overall, these results indicate that simply incentivizing or penalizing the AI, independent of individuals’ performance, use may exacerbate rather than resolve the risk of AI on CI. Thus, we also considered an incentive structure where interventions for AI usage affects the payoffs multiplicatively, thus giving higher payoffs to individuals who rely more (or less) AI and makes better prediction. [SWP: Supp.]

Balanced incentives mitigate AI-induced failures of CI

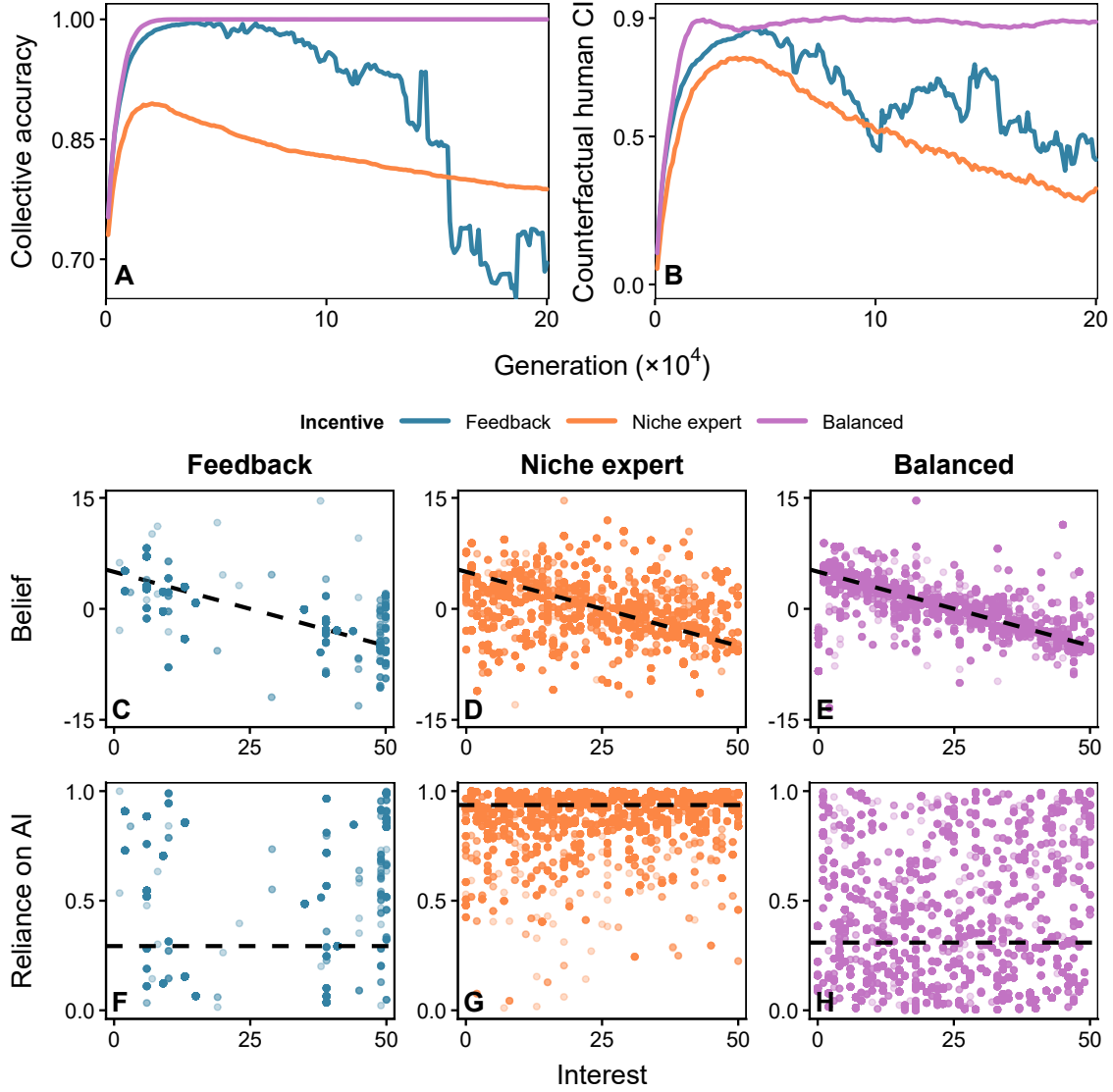


Figure 4: **Balanced structure can promote collective intelligence** Trajectories of collective accuracy (A) and counterfactual human CI (B) under three incentive structures (feedback, niche-expert, and balanced) with Chatbot AI. *[SWP: A and B should have the same y axes scaling.]* Distributions of individual interest and belief (C-E) and of interest and reliance on AI (F-H) after 200,000 generations. Each point represents an individual. Dashed lines in panels C-E represent the true coefficients (α_i) of the prediction target. Dashed horizontal lines in panels F-H indicate the median reliance on AI. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i = 5 - 0.2i$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.4$ and $\tau = 0.3$.

188 Neither the original incentive structures nor simple interventions targeting AI
189 use are sufficient to promote CI. What incentive structure, then, would work in the
190 presence of AI? Here, we propose a “balanced” incentive structure, modeled as the
191 average of the feedback and niche expert incentives.

192 The core idea behind the balanced structure is to maintain interest diversity and
193 prevent collective bias. On one hand, the feedback incentive structure is prone to
194 the loss of interest diversity but can prevent collective bias by incentivizing players
195 whose predictions lie opposite to the collective prediction. On the other hand, the
196 niche-expert structure can maintain interest diversity because it directly incentivizes
197 minority interests. However the niche-expert structure cannot prevent collective bias
198 because, within the same factor of interest, individuals are rewarded only according
199 to their individual predictive performance, which in turn homogenize the predictions.

200 Therefore, when both incentive structures are combined, the population achieves
201 CI by maintaining both interest diversity and limited collective bias (Fig. 4, S3).
202 Compare to the feedback and niche-expert incentive structures, individuals are both
203 evenly distributed through factors (Fig. 4C–E), and their beliefs lie closer to true
204 values (Fig. 4F–H). Moreover, the balanced incentive structure allows for higher lev-
205 els of counterfactual collective accuracy compared to pure feedback and niche-expert
206 scenarios. These patterns are driven by the emergent property of the balanced struc-
207 ture, in which group increases its reliance on AI as the AI becomes more accurate,
208 demonstrating appropriately calibrated AI use.

209 *[SWP: may need some robustness check rather than a case study...]*

210 Discussion

211 The ongoing race towards the development of “better” AIs continues to pose ques-
212 tions about the societal benefits and risks of AI uses. Here, we used mathematical
213 models of evolutionary dynamics to investigate how AI uses and incentive structures
214 jointly affect the emergence of CI, which is defined as the population’s ability to make
215 accurate predictions. We find that an “omniscient” AI model, which incorporates in-
216 formation about all relevant causal factors, can improve collective accuracy, whereas
217 a “chatbot” AI model, whose outcome depends on the user’s interest, can impair it.
218 These contrasting outcomes arise because AI assistance affects not only individual
219 predictions, but also the population’s interest diversity, beliefs, and reliance on AI.
220 We also show that directly incentivizing or penalizing AI uses, independent of in-
221 dividual’s performances, can further limit the collective accuracy of the population.
222 Instead, a balanced incentive structure can maintain CI by promoting diversity and
223 limiting bias.

224 Our analysis reveals that AI can prevent the emergence of CI through two dif-
225 ferent mechanisms. First, AI use can cause individuals to concentrate on a subset of
226 the factors that determine the prediction target, thereby reducing diversity in their
227 interests. This result is broadly consistent with recent empirical studies showing that

228 AI assistance can improve individual performance but limit the diversity of outputs,
229 in domains ranging from scientific research [31] to creative writing [17]. Second, over-
230 reliance on AI can cause the population to inherit inherent biases in AI’s predictions,
231 causing the prediction accuracy to decrease eventually. This mechanism can prevent
232 the emergence of CI even when the interest diversity is maintained because biases in
233 AI predictions can be amplified when individual predictions are aggregated. Thus,
234 preserving the diversity of ideas will not be sufficient to promote CI. Instead, AI
235 uses need to be calibrated to prevent systematic AI biases from becoming collective
236 biases.

237 Our analyses of counterfactual collective accuracy also raises an important concern
238 for future AI uses: while AI uses may improve the apparent performance of
239 individuals and, in some cases, the population as a whole, it can simultaneously reduce
240 the population’s ability to make accurate predictions without AI. These results
241 echo findings from a recent field experiment, which demonstrated that students, who
242 seemingly performed better in solving mathematics problems under AI assistance,
243 performed worse on their own than those who did not have access [32]. These findings
244 highlight the need to evaluate AI tools not only by the performance gains they provide
245 during use, but also by whether individuals and populations retain the ability to
246 perform without them. Making this distinction clear will be particularly important
247 for education, where AI may help students complete a task without helping them
248 learn how to do it on their own.

249 As generative AIs and large language models are already generating substantial
250 benefits, many organizations, both industrial and academic, are increasingly seeking
251 to improve their collective performance by regulating AI uses. Such regulations may
252 take direct forms, including AI-use policies or token-based leaderboards, or may operate
253 indirectly through other incentives, such as productivity rewards or the social stigma
254 associated with AI use. However, our model suggests that directly incentivizing
255 or penalizing AI use is unlikely to induce appropriately calibrated reliance on
256 AI or to promote CI. Incentivizing AI use could lead to overreliance, causing groups
257 to propagate AI biases, whereas penalizing AI use could lead to underreliance, preventing
258 groups from using AI even when it performs well in the relevant domain. Our result
259 instead shows that appropriate AI use and accurate CI can be promoted
260 by focusing on the traditional properties of groups that have been emphasized for
261 decades: maintaining diversity and preventing collective bias, rather than on AI use
262 itself.

263 **[SWP: Older writings:]**

264 Although Omniscient AI can produce group performance that is equal to or better
265 than that achieved without AI, such AI may be difficult to realize in practice
266 because it needs to be able to observe causal factors independently. Achieving this
267 capability would require hardware, such as sensors, as well as autonomous data acquisition
268 capabilities. For this reason, most AI services currently available are better represented
269 by the Chatbot AI in our model, which highlights the potential risks of
270 AI use for CI.

271 In our model, individuals revise their predictions based on the AI’s assistance. It
 272 is therefore unsurprising that AI assistance reduces the diversity of individual pre-
 273 dictions, a pattern that has also been observed empirically. However, a reduction in
 274 prediction diversity does not necessarily lead to the collapse of CI. The principal risk
 275 arises when the incentive structure fails to guide the group toward an appropriate
 276 set of strategies. The feedback structure leads to the breakdown of CI because it
 277 recruits more players to more important factors, which have larger coefficients. This
 278 characteristic creates a risk of interest-diversity loss in finite populations, a situa-
 279 tion in which no player remains interested in certain factors because most players
 280 shift toward relatively important factors. Without AI, this occurs when initial in-
 281 dividual predictions are not sufficiently diverse or when there is no source of novel
 282 ideas, such as mutation. Revising individual predictions with AI assistance exac-
 283 erbates this problem by pulling individuals’ predictions toward the AI’s prediction,
 284 thereby reducing prediction diversity even under the same belief distribution. The
 285 niche-expert structure hinders CI because it favors extreme levels of reliance on AI.
 286 Because individuals within each interest group are affected only by their own perfor-
 287 mance, incentives based on individual performance reach their maximum at either
 288 full reliance on AI or no reliance at all. This extreme reliance results in suboptimal
 289 collective accuracy or substantially delays the emergence of CI.

290 Because AI already generates substantial benefits as well as costs, organizations
 291 increasingly seek to improve collective performance by regulating AI use. Such in-
 292 terventions may take direct forms, including AI-use policies or token-based leader-
 293 boards, or may operate indirectly through other incentives, such as productivity
 294 rewards or the social stigma associated with AI use. However, our model suggests
 295 that directly incentivizing or penalizing AI use is unlikely to induce appropriately
 296 calibrated reliance on AI or to promote CI. Incentivizing AI use could lead to overre-
 297 liance, causing groups to propagate AI biases, whereas penalizing AI use could lead
 298 to underreliance, preventing groups from using AI even when it performs well in the
 299 relevant domain. Our result instead shows that appropriate AI use and accurate CI
 300 can be promoted by focusing on the classic properties of groups that have been em-
 301 phasized for decades—maintaining diversity and preventing collective bias—rather
 302 than on AI use itself.

303 The consistently limited counterfactual collective accuracy observed across AI
 304 types and incentive structures suggests that the presence of AI can degrade hu-
 305 man group’s own performance, consistent with empirical evidence. This may reduce
 306 collective performance when AI assistance becomes unavailable because of resource
 307 constraints, system failure, or model collapse, forcing the group to rebuild CI through
 308 further adaptation. The higher counterfactual collective accuracy achieved under the
 309 balanced structure indicates that an appropriate incentive structure can mitigate the
 310 risk of CI breakdown following the loss of AI assistance.

311 Our model and analysis have several limitations. First, the model considers only
 312 social learning among humans and treats the influence of AI as unidirectional. In
 313 reality, humans can learn from AI, while AI systems can also be further trained

314 on human-generated data. Such bidirectional adaptation may further reduce group
 315 performance by increasing the similarity between human and AI predictions and
 316 thereby weakening their complementarity. Empirical studies of human–AI collabora-
 317 tion likewise suggest that stronger correlations between human and AI predictions
 318 can diminish the benefits of collaboration. Second, we model AI errors using factor-
 319 specific additive biases and random error. Previous studies of AI bias and error
 320 suggest that additive biases may arise from skewed training data, or systematic cal-
 321 ibration. However, AI errors may also exhibit more complex structures, including
 322 biased estimates of factor effects and correlations across prediction errors. Further
 323 empirical research is therefore needed to characterize the structure of AI bias and
 324 error, together with modeling approaches that can incorporate these more complex
 325 patterns. Finally, to retain the tractability of established CI models, we assume that
 326 the prediction target is a linear combination of causal factors, which does not hold
 327 for many real-world tasks. In particular, many machine learning methods used in
 328 AI systems are capable of capturing nonlinear relationships. Future research should
 329 therefore investigate how nonlinear relationships between causal factors and predic-
 330 tion targets affect the interaction between AI assistance and the emergence of CI.

331 Methods

332 Prediction target

333 Following previous models of CI, we model the prediction target as a linear combi-
 334 nation of M independent causal factors:

$$Y = \alpha_0 + \alpha_1 X_1 + \cdots + \alpha_M X_M, \quad (2)$$

335 where the fixed coefficient α_i determines the magnitude and direction of the
 336 effect of factor i , and α_0 is a global intercept. Each causal factor X_i is modeled as an
 337 independent Gaussian random variable with mean zero and standard deviation σ_i .

338 Players

339 We consider a well-mixed population $\mathcal{N}$ consisting of N players engaged in a novel
 340 prediction task. In the absence of AI assistance, the strategy of player A is char-
 341 acterized by two variables: interest e_A and belief c_A . Player A observes the causal
 342 factor X_{e_A} and predicts the target by multiplying the observed factor value by their
 343 belief:

$$Y_A = c_A X_{e_A}, \quad (3)$$

344 where e_A can take an integer value from 0 to M . When $e_A = 0$, the player
 345 does not observe any causal factor and instead makes a constant prediction based
 346 solely on their belief. We refer to this as a stubborn strategy because the player’s

347 prediction does not respond to the realized values of the causal factors. For notational
 348 convenience, we assume that $X_0 = 1$.

349 When the group is assisted by AI, each player’s strategy additionally includes
 350 reliance on AI, denoted by β_A . We assume that each player consults the AI inde-
 351 pendently and revises their prediction by taking a weighted average of their own
 352 prediction and the AI-generated prediction:

$$Y_A^{\text{re}} = \beta_A Y_A^{\text{AI}} + (1 - \beta_A) Y_A, \quad (4)$$

353 where β_A ranges from 0 to 1. A value of $\beta_A = 1$ indicates complete reliance on
 354 AI, whereas $\beta_A = 0$ indicates that the player does not use the AI prediction.

355 AI models

356 We assume that the AI model has been trained on similar tasks and is therefore
 357 capable of making predictions based on causal factors. However, because the model
 358 is imperfect, its outputs are subject to systematic additive bias and stochastic error.
 359 The additive bias associated with factor i is denoted by b_i , and the stochastic error
 360 is denoted by ϵ . The AI is neither trained nor updated during the simulation, and its
 361 biases therefore remain fixed throughout the simulation. Although the bias is indexed
 362 by factor, all b_i , including b_0 , are assigned the same value in the main analyses for
 363 simplicity. The model formulation nevertheless allows biases to differ among factors.

364 We assume that the stochastic error follows a Gaussian distribution with mean
 365 zero and standard deviation τ . Each time a player consults the AI, an indepen-
 366 dent error is generated. Errors are therefore independent across players and across
 367 repeated consultations by the same player. Consequently, stochastic AI errors can
 368 be attenuated when independently generated predictions are aggregated across play-
 369 ers. We therefore held τ constant and varied the magnitude of the additive bias to
 370 manipulate the level of AI accuracy across analyses.

371 We consider two extreme types of AI that differ in the amount of information
 372 available for prediction. The first is termed the Omniscient AI, which can observe
 373 all M causal factors and incorporate them into its prediction regardless of the user’s
 374 interest. The prediction generated by the Omniscient AI for player A is expressed
 375 as follows:

$$Y_A^{\text{AI}} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon. \quad (5)$$

376 The second is termed the Chatbot AI, which can use only the single causal factor
 377 X_{e_A} associated with the player’s interest. The prediction generated by the Chatbot
 378 AI for player A is expressed as follows:

$$Y_A^{\text{AI}} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon. \quad (6)$$

Aggregation rules

The revised individual predictions are aggregated to form a collective prediction. We use two aggregation rules proposed in previous research, depending on the type of AI used by the group.

When the group is assisted by the Omniscient AI, the AI component of each revised prediction contains information about all M factors, even though the individual's own prediction is based on only one factor. Each revised prediction can therefore be treated as an estimate of the entire target. Accordingly, we calculate the collective prediction by taking the simple average of the revised individual predictions:

$$\hat{Y}^{\text{re}} = \frac{Y_1^{\text{re}} + Y_2^{\text{re}} + \cdots + Y_n^{\text{re}}}{N}. \quad (7)$$

When the group is assisted by the Chatbot AI, both the player's initial prediction and the AI's prediction are based on the same causal factor. Each revised prediction therefore contains information about only one factor and can be treated as an estimate of that factor's contribution to the target. We consequently use clustering aggregation.

Clustering aggregation is performed by grouping individuals according to their interests and calculating the mean prediction within each group. The collective prediction is then obtained by summing the mean predictions of all nonempty groups:

$$\hat{Y}^{\text{re}} = \sum_{i:n_i>0} \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{\text{re}}, \quad (8)$$

where $\mathcal{A}_i$ is the set of players interested in factor i , and $n_i = |\mathcal{A}_i|$ is the number of players in that set.

When the group is not assisted by AI, collective predictions are computed from the unrevised individual predictions throughout the simulation. Both simple averaging and clustering can be used in the absence of AI. For fair comparison, we use the same aggregation rule as in the corresponding AI-assisted condition.

Incentive structures

Players receive payoffs determined by an incentive structure based on their strategies. We consider five incentive structures: feedback structure, niche-expert structure, direct intervention on AI use of these two structures, and a balanced structure.

The feedback structure does not directly reward individuals according to their own prediction accuracy. Instead, it rewards predictions that are aligned with the residual error of the collective prediction and therefore have the potential to move the collective prediction toward the true value. Given player A 's revised prediction Y_A^{re} and the collective prediction $\hat{Y}^{\text{re}}$, the payoff is expressed as

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}). \quad (9)$$

412 The niche-expert structure assigns higher payoffs to individuals who make accu-
 413 rate predictions while having an underrepresented interest. Specifically, the squared
 414 prediction error of player A is weighted by $\rho_{e_A} = n_{e_A}/n$, the proportion of players
 415 interested in the same factor:

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2. \quad (10)$$

416 Previous research has shown that these two incentive structures can produce
 417 nearly perfect CI through social learning, except when the niche-expert incentive is
 418 combined with simple averaging. They are therefore used as baseline structures for
 419 examining how AI assistance affects the emergence of CI.

420 To examine the effects of directly encouraging or discouraging AI use, we mod-
 421 ify the feedback and niche-expert structures by adding a term proportional to the
 422 player’s reliance on AI:

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A. \quad (11)$$

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A. \quad (12)$$

423 Here, λ is a weighting coefficient representing the strength of the direct incentive
 424 or penalty applied to AI use. A positive value of λ rewards AI reliance, whereas a
 425 negative value penalizes it. When $\lambda = 0$, no direct incentive or penalty is applied,
 426 and the payoff functions reduce to the original feedback and niche-expert structures.

427 The balanced structure is defined as the arithmetic mean of the feedback and
 428 niche-expert payoff components:

$$\pi_A = \frac{Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - \rho_{e_A}(Y_A^{\text{re}} - Y)^2}{2}. \quad (13)$$

429 Simulations

430 All simulations were conducted using an agent-based model with 50 causal factors
 431 ($M = 50$) and 10,000 players ($n = 10,000$). The parameters defining the prediction
 432 target were sampled at the beginning of the simulations and subsequently held fixed.

433 The coefficients α_i were sampled from a uniform distribution on $[-5, 5]$, except in
 434 the simulation presented in Figure 4, in which α_i was set to $5 - 0.2i$ for visualization.
 435 The standard deviations σ_i were sampled from a uniform distribution on $[0, 3]$.

436 Players’ initial strategies were also sampled from predefined probability distri-
 437 butions. The distribution of individual belief c_A depended on the aggregation rule.
 438 Under simple averaging, each factor’s contribution is scaled by the proportion of
 439 players interested in that factor. We therefore sampled beliefs from a Gaussian dis-
 440 tribution with mean zero and standard deviation 100, following the scaling used
 441 in previous work [13]. Under clustering aggregation, beliefs were sampled from a
 442 Gaussian distribution with mean zero and standard deviation 5. Because predictions

are averaged separately within each interest group and then summed, no additional scaling of c_A is required. The initial interest of each player was sampled uniformly from the integers between 0 and 50, and reliance on AI was sampled from a uniform distribution on $[0, 1]$.

We assumed that prediction events occurred on a substantially faster timescale than social learning. At each generation, we therefore calculated the expected payoff $\mathbb{E}[\pi_A]$ of each player analytically rather than explicitly sampling prediction events. The expectation was taken over the distributions of the causal factors $X_1, \dots, X_M$ and the stochastic AI error ϵ , conditional on the current population strategies and the fixed target parameters. Closed form expression of expected payoffs were derived for all incentive structures and are presented in the Supplementary Methods.

Each generation consisted of an expected payoff calculation followed by one opportunity for pairwise social learning. Two distinct players, A and B , were randomly sampled, and player A imitated all three strategies of player B with probability

$$\frac{1}{1 + \exp(s(\mathbb{E}[\pi_A] - \mathbb{E}[\pi_B]))}, \quad (14)$$

where s represents the strength of social learning and was set to 50 in all simulations. We introduced a mutation rate μ , such that with probability μ , player A did not imitate player B but instead redrew all strategy components from the same distributions used to initialize the population. The mutation rate was set to zero in all simulations except those presented in Supplementary Figure 4.

Unless otherwise specified, each simulation was run for 200,000 generations. When a result was represented by a single summary value rather than as a temporal trajectory, the metric was averaged over the final 10,000 generations.

To evaluate the dynamics of collective prediction, we defined six summary metrics: collective accuracy, collective bias, collective variance, counterfactual collective accuracy, interest diversity, and median reliance on AI. At each population state, expectations and variances were evaluated analytically over the distributions of the causal factors and AI errors rather than estimated from sampled prediction events. Collective bias and collective variance constitute the bias–variance decomposition of the group level collective prediction error.

Collective accuracy was normalized relative to the baseline prediction α_0 . A negative value indicates that the collective prediction performs worse than the baseline. Counterfactual collective accuracy was evaluated by setting $\beta_A = 0$ while retaining the evolved interests and beliefs of the AI-assisted population. In calculating interest diversity, we used the convention that $\rho_i \ln \rho_i = 0$ when $\rho_i = 0$ (Table 1).

Data and code availability

All data are stored in a publicly available GitHub repository (<https://github.com/parklab-snu/collective-intelligence-with-AI>).

Table 1: Evaluation metrics used to analyze collective prediction dynamics.

Metric	Definition	Interpretation
Collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y}^{\text{re}})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy of the revised collective prediction relative to the baseline prediction α_0 .
Collective bias	$\left(\mathbb{E}[Y - \hat{Y}^{\text{re}}]\right)^2$	Squared expected deviation of the revised collective prediction from the true target.
Collective variance	$\text{Var}(Y - \hat{Y}^{\text{re}})$	Variability of the revised collective prediction error across prediction events.
Counterfactual collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy obtained after removing AI use from the current population while retaining its evolved interests and beliefs.
Interest diversity	$\exp\left(-\sum_{i=0}^M \rho_i \ln \rho_i\right)$	Effective number of interests represented in the population.
Median reliance on AI	$\text{median}_{A \in N}(\beta_A)$	Median reliance on AI in the population.

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484 **Contributions**

485 Jaeseung Kim: Conceptualization, Formal Analysis, Investigation, Methodology,
486 Software, Visualization, Writing - Original Draft Preparation, Writing - Review &
487 Editing. Sang Woo Park: Conceptualization, Investigation, Methodology, Supervi-
488 sion, Writing - Original Draft Preparation, Writing - Review & Editing.

489 **Competing Interests**

490 The authors declare no competing interests.

Supplementary Text

1 Omniscient AI

Omniscient AI assumes that AI can incorporate all causal factors regardless of the player's interest.

1.1 Model

$$Y_A^{AI} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon. \quad (S1)$$

For simplicity, prediction of Omniscient AI can be written as:

$$Y^{AI} = Y + B + \epsilon \quad (S2)$$

where Y stands for true prediction target ($Y = \alpha_0 + \alpha_1 X_1 + \cdots + \alpha_M X_M$) and B is sum of all fixed biases ($B = b_0 + b_1 + \cdots + b_M$), ϵ follows normal distribution with zero mean, τ standard deviation with no correlation within or between players. Player A's prediction and revised prediction can be written as follows:

$$Y_A = c_A X_{e_A} \quad (S3)$$

$$Y_A^{re} = \beta_A Y^{AI} + (1 - \beta_A) Y_A \quad (S4)$$

where c_A stands for player A's belief on the factor, e_A stands for the interest, and β_A stands for reliance on AI.

1.2 Aggregation

Because Y_A^{AI} is subject to all M factors, we use simple averaging aggregation for collective prediction.

$$\begin{aligned} \hat{Y}^{re} &= \frac{1}{N} \sum_{j=1}^N Y_j^{re} \\ &= \frac{1}{N} \sum_{j=1}^N \{ \beta_j Y^{AI} + (1 - \beta_j) Y_j \} \\ &= \frac{1}{N} \sum_{j=1}^N \{ \beta_j (Y + B + \epsilon) + (1 - \beta_j) c_j X_{e_j} \} \end{aligned}$$

Reparameterization for simple derivation and interpretability.

$$\bar{\beta} := \frac{1}{N} \sum_{j=1}^N \beta_j \quad (S5)$$

508

$$\bar{\eta} := \frac{1}{N} \sum_{j=1}^N \beta_j \epsilon \quad (\text{S6})$$

509

$$\gamma_i := \frac{1}{N} \sum_{A \in \mathcal{A}_i} (1 - \beta_A) c_A \quad (\text{S7})$$

510 $\bar{\beta}$ can be interpreted as mean reliance on AI, γ stands for the coefficient for the factor
 511 i which is solely based on players' belief.

512 By this reparameterization, we can write $\hat{Y}^{re}$ as follows.

$$\hat{Y}^{re} = \bar{\beta}(Y + B) + \sum_{i=0}^M \gamma_i X_i + \bar{\eta} \quad (\text{S8})$$

513 1.3 Expected payoffs

514 Feedback

$$\delta_0 := \alpha_0 - (\bar{\beta}(\alpha_0 + B) + \gamma_0) \quad (\text{S9})$$

515

$$\delta_i := \alpha_i - (\bar{\beta}\alpha_i + \gamma_i), \quad (i \neq 0) \quad (\text{S10})$$

516

$$Y - \hat{Y}^{re} = \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \quad (\text{S11})$$

517 See equation Eq. (S6) for the $\bar{\eta}$.

518 δ_0 is constant term and δ_i is independent of α_k ($k \neq i$), ϵ , and $\bar{\eta}$.

$$\begin{aligned} \mathbb{E}(\pi_A) &= \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] \\ &= \mathbb{E} \left[\left\{ \beta_A \alpha_0 + \beta_A \sum_{i=1}^M \alpha_i X_i + \beta_A B + \beta_A \epsilon + (1 - \beta_A) c_A X_{e_A} \right\} \left\{ \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \right\} \right] \\ &= \beta_A \alpha_0 \delta_0 + \beta_A \sum_{i=1}^M \delta_i \alpha_i \sigma_i^2 + \beta_A B \delta_0 - \frac{1}{N} \beta_A^2 \tau^2 + (1 - \beta_A) c_A \delta_{e_A} \sigma_{e_A}^2 \\ &= \beta_A \left\{ (\alpha_0 + B) \delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2 \right\} + (1 - \beta_A) c_A \delta_{e_A} \sigma_{e_A}^2 - \frac{\beta_A^2 \tau^2}{N} \end{aligned}$$

519 **Niche-expert**

520 If $e_A = 0$, player does not observe causal factor and their unrevised prediction is same
 521 with their belief c_A . Such that, the expected payoff is

$$\begin{aligned}
 \mathbb{E}(\pi_A) &= -\mathbb{E}[\rho_0(Y_A^{re} - Y)^2] \\
 &= -\mathbb{E}[\rho_0\{(\beta_A - 1)Y + \beta_A B + \beta_A \epsilon + (1 - \beta_A)c_A\}^2] \\
 &= -\mathbb{E}\left[\rho_0\left\{(\beta_A - 1)(\alpha_0 - c_A) + \beta_A B + (\beta_A - 1)\sum_{i=1}^M \alpha_i X_i + \beta_A \epsilon\right\}^2\right] \\
 &= -\rho_0\left\{(1 - \beta_A)^2(\alpha_0 - c_A)^2 - 2\beta_A(1 - \beta_A)(\alpha_0 - c_A)B + \beta_A^2 B^2\right. \\
 &\quad \left.+ (1 - \beta_A)^2 \sum_{i \neq 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2\right\}
 \end{aligned}$$

522 Else,

$$\begin{aligned}
 \mathbb{E}(\pi_A) &= -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2] \\
 &= -\mathbb{E}[\rho_{e_A}\{(\beta_A - 1)Y + \beta_A B + \beta_A \epsilon + (1 - \beta_A)c_A X_{e_A}\}^2] \\
 &= -\mathbb{E}\left[\rho_{e_A}\left\{(\beta_A - 1)\alpha_0 + \beta_A B + (\beta_A - 1)(\alpha_{e_A} - c_A)X_{e_A} + (\beta_A - 1)\sum_{i \neq e_A, 0} \alpha_i X_i + \beta_A \epsilon\right\}^2\right] \\
 &= -\rho_{e_A}\left\{(1 - \beta_A)^2 \alpha_0^2 - 2\beta_A(1 - \beta_A)\alpha_0 B + \beta_A^2 B^2 + (1 - \beta_A)^2(\alpha_{e_A} - c_A)^2 \sigma_{e_A}^2\right. \\
 &\quad \left.+ (1 - \beta_A)^2 \sum_{i \neq e_A, 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2\right\}
 \end{aligned}$$

523 **Incentivizing or penalizing AI use**

524 Because additive term $\lambda\beta$ is not a random variable, we can simply derive the expected
 525 payoff as

$$\mathbb{E}(\pi_{A, Feedback}) = \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re}) + \lambda\beta_A] = \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] + \lambda\beta_A. \quad (S12)$$

$$\mathbb{E}(\pi_{A, Niche-expert}) = -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2 + \lambda\beta_A] = -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2] + \lambda\beta_A. \quad (S13)$$

526 **1.4 Expected accuracy**

$$\mathbb{E}[(Y - \alpha_0)^2] = \mathbb{E}\left[\left(\alpha_0 + \sum_{i=1}^M \alpha_i X_i - \alpha_0\right)^2\right] = \sum_{i=1}^M \alpha_i^2 \sigma_i^2 \quad (S14)$$

527

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] = \mathbb{E}\left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i - \hat{\eta}\right)^2\right] = \delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{j=1}^N \beta_j^2 \quad (\text{S15})$$

528

$$\begin{aligned} R &= 1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]} \\ &= 1 - \frac{\delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{j=1}^N \beta_j^2}{\sum_{i=1}^M \alpha_i^2 \sigma_i^2} \end{aligned}$$

529 2 Chatbot AI

530 2.1 Model

$$Y_A^{AI} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon \quad (\text{S16})$$

531 b_{e_A} stands for factor specific additive bias, ϵ follows normal distribution with zero
 532 mean, τ standard deviation with no correlation within or between players. Player
 533 A's prediction and revised prediction can be written as follows:

$$Y_A = c_A X_{e_A} \quad (\text{S17})$$

534

$$Y_A^{re} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A \quad (\text{S18})$$

535 where c_A stands for player A's belief on the factor, e_A stands for the interest, and
 536 β_A stands for reliance on AI.

537 2.2 Aggregation

538 For Chatbot AI, clustering aggregation is used for collective prediction.

$$\begin{aligned} \hat{Y}^{re} &= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{re} \\ &= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \{\beta_A Y_A^{AI} + (1 - \beta_A) Y_A\} \\ &= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} [\{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon] \end{aligned}$$

539 Reparameterize for the simplicity.

$$\tilde{c}_i := \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} \quad (\text{S19})$$

540

$$\tilde{B}_i := \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A b_{e_A} \quad (\text{S20})$$

541

$$\bar{B} := \sum_{i=0}^M \tilde{B}_i \quad (\text{S21})$$

542

$$\tilde{\eta}_i := \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A \epsilon \quad (\text{S22})$$

543

$$\bar{\eta} := \sum_{i=0}^M \tilde{\eta}_i \quad (\text{S23})$$

544 2.3 Expected payoffs

545 Feedback

546

$$\delta_0 := \alpha_0 - \tilde{c}_0 - \bar{B} \quad (\text{S24})$$

547

$$\delta_i := \alpha_i - \tilde{c}_i, \quad (i \neq 0) \quad (\text{S25})$$

$$Y - \hat{Y}^{re} = \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \quad (\text{S26})$$

548 δ_0 is constant term and δ_i is independent of α_k ($k \neq i$), ϵ , and $\bar{\eta}$.

$$\begin{aligned} \mathbb{E}(\pi_A) &= \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] \\ &= \mathbb{E} \left[\{(\beta_A \alpha_{e_A} + (1 - \beta_A) c_A) X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon\} \left\{ \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \right\} \right] \\ &= \{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} \delta_{e_A} \sigma_{e_A}^2 + \beta_A b_{e_A} \delta_0 - \frac{1}{n_{e_A}} \beta_A^2 \tau^2 \end{aligned}$$

549 Niche-expert

550 If $e_A = 0$,

$$\begin{aligned} \mathbb{E}(\pi_A) &= -\mathbb{E}[\rho_0(Y_A^{re} - Y)^2] \\ &= -\mathbb{E} \left[\rho_0 \left[\{\beta_A \alpha_0 + (1 - \beta_A) c_A\} + \beta_A b_0 + \beta_A \epsilon - \alpha_0 - \sum_{i=1}^M \alpha_i X_i \right]^2 \right] \\ &= -\rho_0 \left\{ (1 - \beta_A)^2 (c_A - \alpha_0)^2 + 2(1 - \beta_A)(c_A - \alpha_0) \beta_A b_0 + \beta_A^2 b_0^2 \right. \\ &\quad \left. + \sum_{i \neq 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2 \right\} \end{aligned}$$

551 Else,

$$\begin{aligned}
\mathbb{E}(\pi_A) &= -\mathbb{E}[\rho_{e_A}(Y_A^{\text{re}} - Y)^2] \\
&= -\mathbb{E}\left[\rho_{e_A}\left[\{\beta_A\alpha_{e_A} + (1 - \beta_A)c_A\}X_{e_A} + \beta_A b_{e_A} + \beta_A\epsilon - \alpha_0 - \sum_{i=1}^M \alpha_i X_i\right]^2\right] \\
&= -\rho_{e_A}\left\{(1 - \beta_A)^2(c_A - \alpha_{e_A})^2\sigma_{e_A}^2 + \sum_{i \neq e_A, 0} \alpha_i^2 \sigma_i^2\right. \\
&\quad \left.+ \alpha_0^2 + \beta_A^2 b_{e_A}^2 - 2\alpha_0\beta_A b_{e_A} + \beta_A^2 \tau^2\right\}
\end{aligned}$$

552 Incentivizing or penalizing AI use

553 Because additive term $\lambda\beta$ is not a random variable, we can simply derive the expected
554 payoff as

$$\mathbb{E}(\pi_{A, \text{Feedback}}) = \mathbb{E}\left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A\right] = \mathbb{E}\left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}})\right] + \lambda\beta_A. \quad (\text{S27})$$

$$\mathbb{E}(\pi_{A, \text{Niche-expert}}) = -\mathbb{E}\left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A\right] = -\mathbb{E}\left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2\right] + \lambda\beta_A. \quad (\text{S28})$$

555 Balanced

556 Expected payoff of the balanced structure can also be driven by arithmetic mean of
557 expected payoff of feedback and niche-expert.

$$\mathbb{E}(\pi_A) = \mathbb{E}\left[\frac{Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - \rho_{e_A}(Y_A^{\text{re}} - Y)^2}{2}\right] = \frac{\mathbb{E}\left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}})\right] - \mathbb{E}\left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2\right]}{2} \quad (\text{S29})$$

558 2.4 Expected accuracy

$$\mathbb{E}[(Y - \alpha_0)^2] = \mathbb{E}\left[\left(\alpha_0 + \sum_{i=1}^M \alpha_i X_i - \alpha_0\right)^2\right] = \sum_{i=1}^M \alpha_i^2 \sigma_i^2 \quad (\text{S30})$$

$$\mathbb{E}[(Y - \hat{Y})^2] = \mathbb{E}\left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i - \hat{\eta}\right)^2\right] = \delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \sum_{i=0}^M \frac{\tau^2}{n_i^2} \sum_{A \in \mathcal{A}_i} \beta_A^2 \quad (\text{S31})$$

$$\begin{aligned}
R &= 1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]} \\
&= 1 - \frac{\delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \sum_{i=0}^M \frac{\tau^2}{n_i^2} \sum_{A \in \mathcal{A}_i} \beta_A^2}{\sum_{i=1}^M \alpha_i^2 \sigma_i^2}
\end{aligned}$$

561 3 Evolutionary dynamics

562 For sufficiently large population ($N \rightarrow \infty$), we can approximate the deterministic
563 evolutionary dynamics using the replicator equation. Let λ denote a strategy, p_λ its
564 frequency, and $u_\lambda = \mathbb{E}(\pi_\lambda)$ its expected payoff. The replicator equation is

$$\frac{dp_\lambda}{dt} = p_\lambda(u_\lambda - \bar{u}), \quad (\text{S32})$$

565 where $\bar{u} = \sum_\lambda p_\lambda u_\lambda$. Therefore, at an internal stationary state, all strategies
566 with positive frequency have the same expected payoff. A boundary state can also
567 be stationary because a strategy with exactly zero frequency cannot increase under
568 the replicator equation. Thus, stability of a boundary state needs to be considered
569 separately by examining invasion by a rare strategy.

570 3.1 Omniscient AI

571 3.1.1 Feedback

572 When the population size approaches infinity, the effect of stochastic error on the
573 collective prediction vanishes ($\bar{\eta} \rightarrow 0$) and the effect of stochastic error in the ex-
574 pected payoff, $-\beta^2 \tau^2 / N$, also vanishes. Following prior research [13], we define the
575 Lyapunov function as the expected error of collective prediction

$$V = \mathbb{E} \left[(Y - \hat{Y}^{re})^2 \right]. \quad (\text{S33})$$

576 The function V is non-negative and equals zero when the collective prediction is
577 identical to the target. To examine how V changes under the evolutionary dynamics,
578 we take its time derivative:

$$\begin{aligned}
\frac{dV}{dt} &= -2\mathbb{E} \left[(Y - \hat{Y}^{re}) \frac{d\hat{Y}^{re}}{dt} \right] \\
&= -2\mathbb{E} \left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i \right) \left\{ \frac{d\bar{\beta}}{dt} (Y + B) + \sum_{i=0}^M \frac{d\gamma_i}{dt} X_i \right\} \right] \\
&= -2 \left[\left\{ (\alpha_0 + B) \delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2 \right\} \frac{d\bar{\beta}}{dt} + \delta_0 \frac{d\gamma_0}{dt} + \sum_{i=1}^M \delta_i \sigma_i^2 \frac{d\gamma_i}{dt} \right].
\end{aligned}$$

579 For notational convenience, we define

$$K = (\alpha_0 + B) \delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2. \quad (\text{S34})$$

580 Although K changes with the population state, it is common to all strategies at a
581 given state and can therefore be treated as a constant with respect to the strategy
582 index λ . We also define

$$G_e = \begin{cases} \delta_0, & e = 0, \\ \delta_e \sigma_e^2, & e \neq 0. \end{cases} \quad (\text{S35})$$

583 By the definitions of $\bar{\beta}$ and γ_i and the replicator equation,

$$\frac{d\bar{\beta}}{dt} = \sum_{\lambda} p_{\lambda} (u_{\lambda} - \bar{u}) \beta_{\lambda} = \text{Cov}(\beta, u). \quad (\text{S36})$$

584 Similarly, the terms involving γ_i can be combined as

$$\delta_0 \frac{d\gamma_0}{dt} + \sum_{i=1}^M \delta_i \sigma_i^2 \frac{d\gamma_i}{dt} = \text{Cov}((1 - \beta) c G_e, u). \quad (\text{S37})$$

585 Therefore,

$$\begin{aligned}
\frac{dV}{dt} &= -2 \{ K \text{Cov}(\beta, u) + \text{Cov}((1 - \beta) c G_e, u) \} \\
&= -2 \text{Cov}(K \beta + (1 - \beta) c G_e, u).
\end{aligned}$$

586 For the feedback structure in the limit $N \rightarrow \infty$, the expected payoff of strategy
587 λ is

$$u_{\lambda} = K \beta_{\lambda} + (1 - \beta_{\lambda}) c_{\lambda} G_{e_{\lambda}}. \quad (\text{S38})$$

588 Which is same with Thus,

$$\frac{dV}{dt} = -2 \text{Cov}(u, u) = -2 \text{Var}(u) \leq 0. \quad (\text{S39})$$

Therefore, the expected error of collective prediction cannot increase during the evolutionary process and decreases whenever strategies differ in their expected payoffs. The derivative becomes zero when the variance of expected payoffs is zero, $\text{Var}(u) = 0$. This condition corresponds to equal expected payoffs among the strategies present in the population. A boundary state in which the entire population consists of a single strategy can also satisfy $\text{Var}(u) = 0$ even when $V > 0$. However, such a state is not evolutionarily stable under the feedback structure. If the collective prediction differs from the target Y , a nonzero residual remains. By the definition of the feedback payoff, a novel strategy whose prediction aligns with the direction of this residual obtains a higher expected payoff than the resident strategy and can therefore invade the population. Thus, an imperfect boundary equilibrium with $V > 0$ is unstable to the introduction of novel predictions.

Consequently, the evolutionary dynamics continue to reduce the collective prediction error until the population reaches the stable state with zero residual, $V = 0$. At this state, the collective prediction equals the target, $\hat{Y}^{re} = Y$, and the collective accuracy reaches $R = 1$.

3.1.2 Niche-expert

Unlike the stochastic error in the collective prediction, stochastic error in the niche-expert payoff is evaluated at the individual level and does not vanish as population size increases. Therefore, we retain $\tau > 0$ in the following derivation.

We first consider strategies having the same interest by optimizing the expected payoff with respect to c and β . For a nonzero interest e , the expected payoff is strictly concave with respect to c for $\beta < 1$ and $\sigma_e > 0$, and is maximized at

$$c_e^* = \alpha_e. \quad (\text{S40})$$

After substituting $c_e^* = \alpha_e$, the payoff is strictly concave with respect to β . Its unconstrained stationary point is

$$\tilde{\beta}_e = \frac{\alpha_0^2 + \alpha_0 B + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2}{(\alpha_0 + B)^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S41})$$

Because β is constrained between 0 and 1, the optimal β_e^* is $\tilde{\beta}_e$ when $0 < \tilde{\beta}_e < 1$, and otherwise lies at the corresponding boundary, $\beta_e^* = 0$ or $\beta_e^* = 1$. Thus, the boundary cases do not require a separate evolutionary argument; they only change the optimal prediction and its minimum expected squared error. When $\beta_e^* = 1$, the revised prediction is independent of c_e , so different values of c_e are selectively neutral and can be treated as a single payoff-equivalent strategy class.

Let L_e denote the minimum expected individual error, $\mathbb{E}[(Y_A^{re} - Y)^2]$, within interest e . For an internal optimum,

$$L_e = \frac{B^2 \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2 \left(\alpha_0^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 \right)}{(\alpha_0 + B)^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S42})$$

622 If the optimum is located at $\beta_e^* = 0$ or $\beta_e^* = 1$, L_e is instead obtained by evaluating
 623 the expected squared prediction error at the corresponding boundary.

624 For interest 0, optimization with respect to c gives

$$c_0^*(\beta) = \alpha_0 - \frac{\beta}{1 - \beta} B. \quad (\text{S43})$$

625 Substituting this into the expected payoff gives a strictly concave function of β ,
 626 with the unique optimum

$$\beta_0^* = \frac{\sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S44})$$

627 Therefore,

$$c_0^* = \alpha_0 - \frac{B}{\tau^2} \sum_{j=1}^M \alpha_j^2 \sigma_j^2, \quad (\text{S45})$$

628 and

$$L_0 = \frac{\tau^2 \sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S46})$$

629 We denote the optimal strategy, or the payoff-equivalent strategy class when
 630 $\beta_e^* = 1$, by S_e . A state containing a suboptimal strategy within an interest can
 631 be stationary if the optimal strategy is absent, but it is unstable against invasion
 632 because the optimal strategy has a strictly higher payoff at the same ρ_e . Therefore,
 633 a stable state can contain only S_e within each represented interest.

634 We next consider the frequencies of the $M + 1$ optimal strategies. Let ρ_e denote
 635 the frequency of S_e . Its expected payoff is

$$u_e = -\rho_e L_e. \quad (\text{S47})$$

636 Assuming $L_e > 0$ for all interests, the equal-payoff condition of the replicator
 637 equation at an internal stationary state gives

$$\rho_0^* L_0 = \rho_1^* L_1 = \cdots = \rho_M^* L_M. \quad (\text{S48})$$

638 Together with $\sum_{e=0}^M \rho_e^* = 1$, this uniquely determines

$$\rho_e^* = \frac{D}{L_e}, \quad D = \left(\sum_{j=0}^M \frac{1}{L_j} \right)^{-1}. \quad (\text{S49})$$

639 This internal stationary state is locally stable. At equilibrium, all represented
 640 interests have the same expected payoff, with $\rho_e^* L_e = D$. If the frequency of one
 641 interest increases slightly above its equilibrium value, its payoff decreases because
 642 the niche-expert payoff is proportional to $-\rho_e L_e$. Conversely, if the frequency of
 643 an interest decreases below its equilibrium value, its payoff increases. The replicator
 644 dynamics therefore generate a restoring force that reduces small deviations in interest
 645 frequencies and drives the population back toward the internal stationary state.

646 If one or more interests are absent, the resulting boundary state can be stationary
 647 under the replicator equation, but it is unstable against invasion. The payoff of a rare
 648 absent interest approaches zero as $\rho_e \rightarrow 0$, whereas the mean payoff of the resident
 649 population is negative when $L_e > 0$, so the rare interest increases when introduced
 650 at sufficiently small frequency.

651 Thus, there exists a unique stable full-support equilibrium in terms of the optimal
 652 prediction strategies and their frequencies. Whether the constrained optimum has
 653 $\beta_e^* = 0$, $0 < \beta_e^* < 1$, or $\beta_e^* = 1$ does not change the frequency argument, because
 654 these cases differ only through the resulting value of L_e . When $\beta_e^* = 1$, the strategy
 655 is selectively neutral with respect to c_e , but its prediction, payoff, and equilibrium
 656 interest frequency remain uniquely determined.

657 This equilibrium also allows a direct comparison with the expected accuracy of
 658 the AI. Using Y^{AI} itself without aggregation as the prediction gives

$$\mathbb{E} [(Y - Y^{\text{AI}})^2] = B^2 + \tau^2. \quad (\text{S50})$$

659 For every interest, complete reliance on AI, $\beta = 1$, is a feasible strategy. There-
 660 fore, because L_e is the minimum individual expected squared prediction error within
 661 interest e ,

$$L_e \leq B^2 + \tau^2. \quad (\text{S51})$$

662 Moreover, for interest 0,

$$L_0 = \frac{\tau^2 \sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2} < \tau^2 \leq B^2 + \tau^2. \quad (\text{S52})$$

663 Because the stable equilibrium has $\rho_0^* > 0$,

$$\sum_{e=0}^M \rho_e^* L_e < B^2 + \tau^2. \quad (\text{S53})$$

664 The collective prediction is the average of individual revised predictions. By
 665 convexity of the squared error,

$$\mathbb{E} [(Y - \hat{Y}^{re})^2] \leq \sum_{e=0}^M \rho_e^* L_e. \quad (\text{S54})$$

Therefore,

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] < \mathbb{E}[(Y - Y^{AI})^2]. \quad (\text{S55})$$

Because the collective accuracy and AI accuracy use the same normalization by $\mathbb{E}[(Y - \alpha_0)^2]$, the stationary collective accuracy under the niche-expert structure is strictly greater than the expected accuracy of the AI prediction.

Although the niche-expert structure improves collective prediction relative to the AI, the stationary collective accuracy does not reach one when $\tau > 0$. For a nonzero factor i , the residual coefficient of the stationary collective prediction can be written as

$$\delta_i^* = \alpha_i \{1 - \bar{\beta}^* - \rho_i^*(1 - \beta_i^*)\} = \alpha_i \sum_{e \neq i} \rho_e^*(1 - \beta_e^*). \quad (\text{S56})$$

At the stable equilibrium, $\rho_0^* > 0$. In addition, $\tau > 0$ implies $\beta_0^* < 1$. Therefore, for every factor with $\alpha_i \neq 0$,

$$\delta_i^* \neq 0. \quad (\text{S57})$$

Because $\sum_{i=1}^M \alpha_i^2 \sigma_i^2 > 0$, at least one factor contributes positive prediction error. Hence,

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] > 0, \quad (\text{S58})$$

and consequently the stationary collective accuracy satisfies $R^* < 1$.

3.2 Chatbot AI

3.2.1 Feedback

With Chatbot AI, the effect of stochastic error on the collective prediction depends on n_i , requiring the additional assumption that $n_i \rightarrow \infty$ for every interest for the stochastic error to vanish. However, Under this assumption, the loss of interest diversity cannot occur, and therefore the stochastic breakdown of collective accuracy observed in finite populations cannot be analytically captured. Instead, by assuming that each interest is represented by infinitely many players, we can show that the expected collective prediction error monotonically decreases until perfect accuracy is achieved. Because finite population dynamics are more relevant to the empirical populations considered in this study, we retain finite population simulations for the main result.

3.2.2 Niche-expert

We first consider strategies having the same interest by optimizing the expected payoff with respect to c and β . For a nonzero interest e , the expected payoff is strictly concave with respect to c for $\beta < 1$ and $\sigma_e > 0$, and is maximized at

$$c_e^* = \alpha_e. \quad (\text{S59})$$

695 After substituting $c_e^* = \alpha_e$, the payoff is strictly concave with respect to β . Its
 696 unconstrained stationary point is

$$\tilde{\beta}_e = \frac{2\alpha_0 b_e}{2b_e^2 + 2\tau^2}. \quad (\text{S60})$$

697 Because β is constrained between 0 and 1, the optimal β_e^* is $\tilde{\beta}_e$ when $0 < \tilde{\beta}_e < 1$,
 698 and otherwise lies at the corresponding boundary, $\beta_e^* = 0$ or $\beta_e^* = 1$. Because we
 699 assumes AI model with moderate accuracy, b_e and τ is relatively smaller than α_0
 700 which makes $\tilde{\beta}_e > 1$ when $\alpha_0 b_e > 0$ and $\tilde{\beta}_e < 0$ when $\alpha_0 b_e < 0$. These boundary
 701 cases lead to extreme reliance on AI or complete exclusion of AI which both hinders
 702 the emergence of CI. Only when the prediction task is centered near zero and AI is
 703 relatively inaccurate ($\alpha_0 b_e < b_e^2 + \tau^2$), and $\alpha_0 b_e > 0$, the internal optimum β_e^* can be
 704 exist.

705 For interest 0, optimization with respect to c gives

$$c_0^*(\beta) = \alpha_0 - \frac{\beta}{1 - \beta} b_0. \quad (\text{S61})$$

706 Substituting this into the expected payoff gives a strictly concave function of β , with
 707 the unique optimum

$$\beta_0^* = 0. \quad (\text{S62})$$

708 **Supplementary Figure**

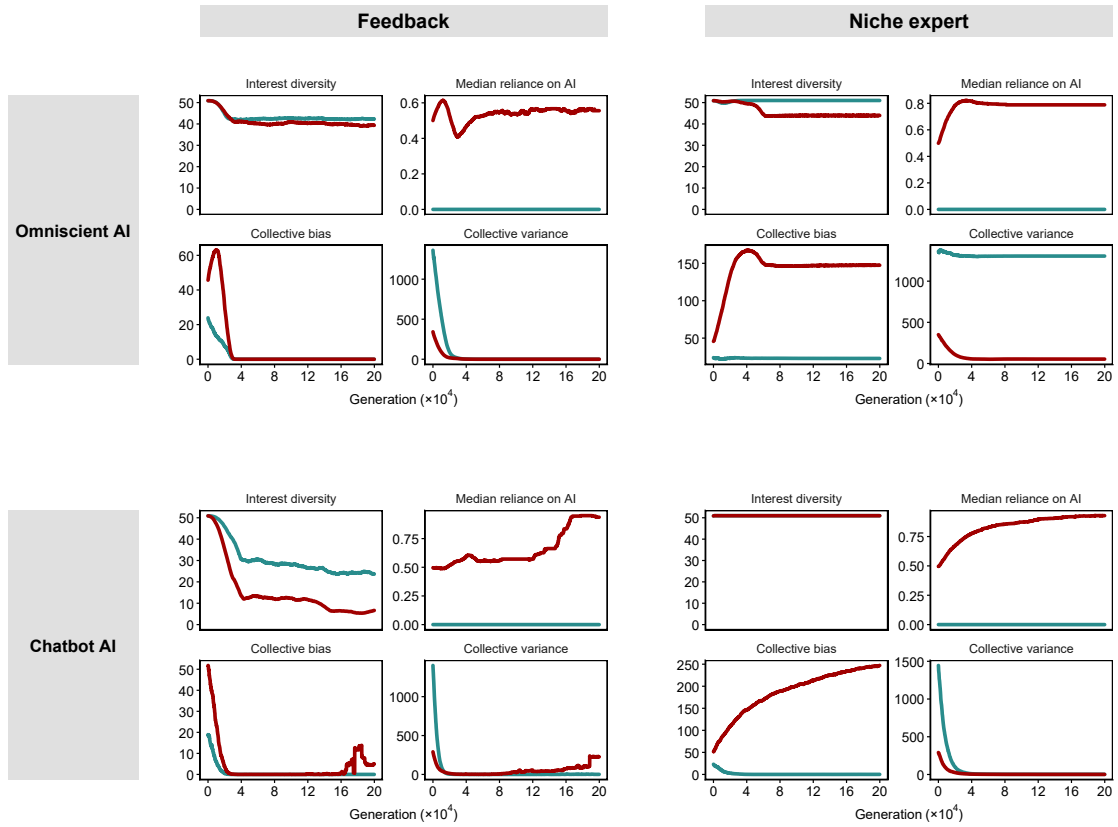


Figure S1: **Caption.** Text.

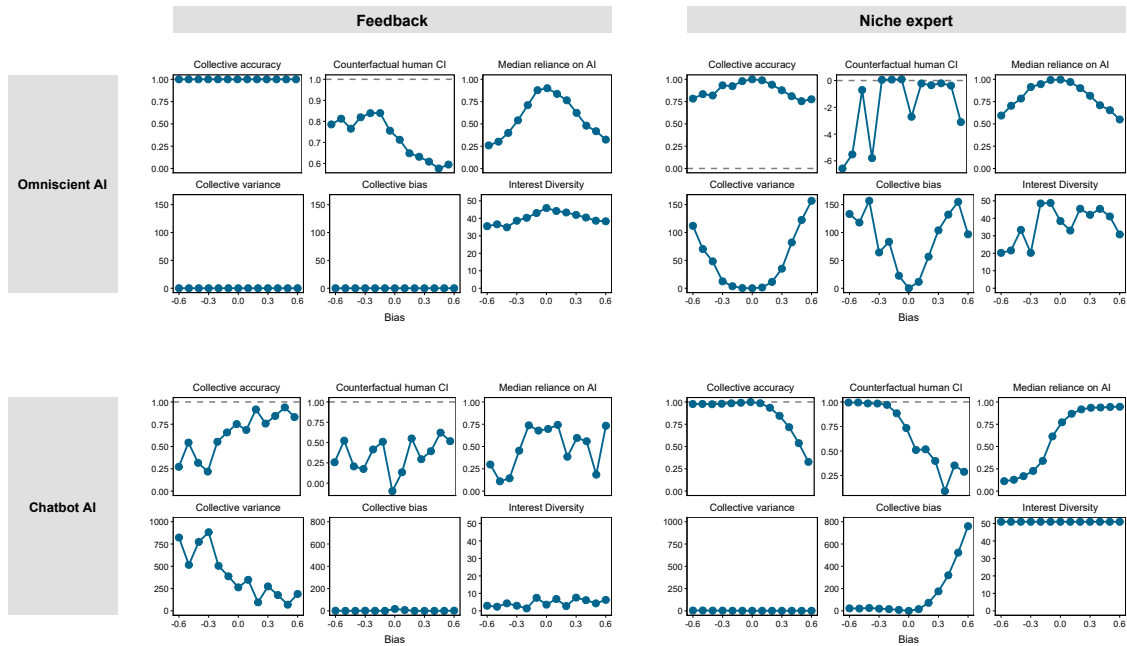


Figure S2: Caption. Text.

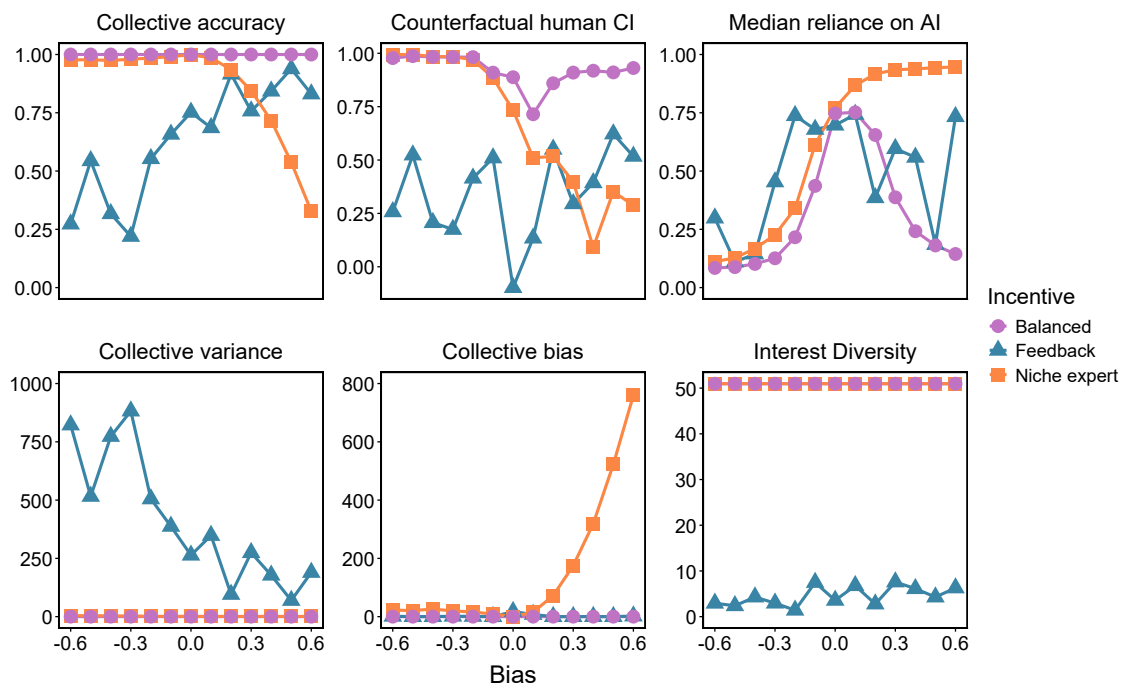


Figure S3: **Caption.** Text.

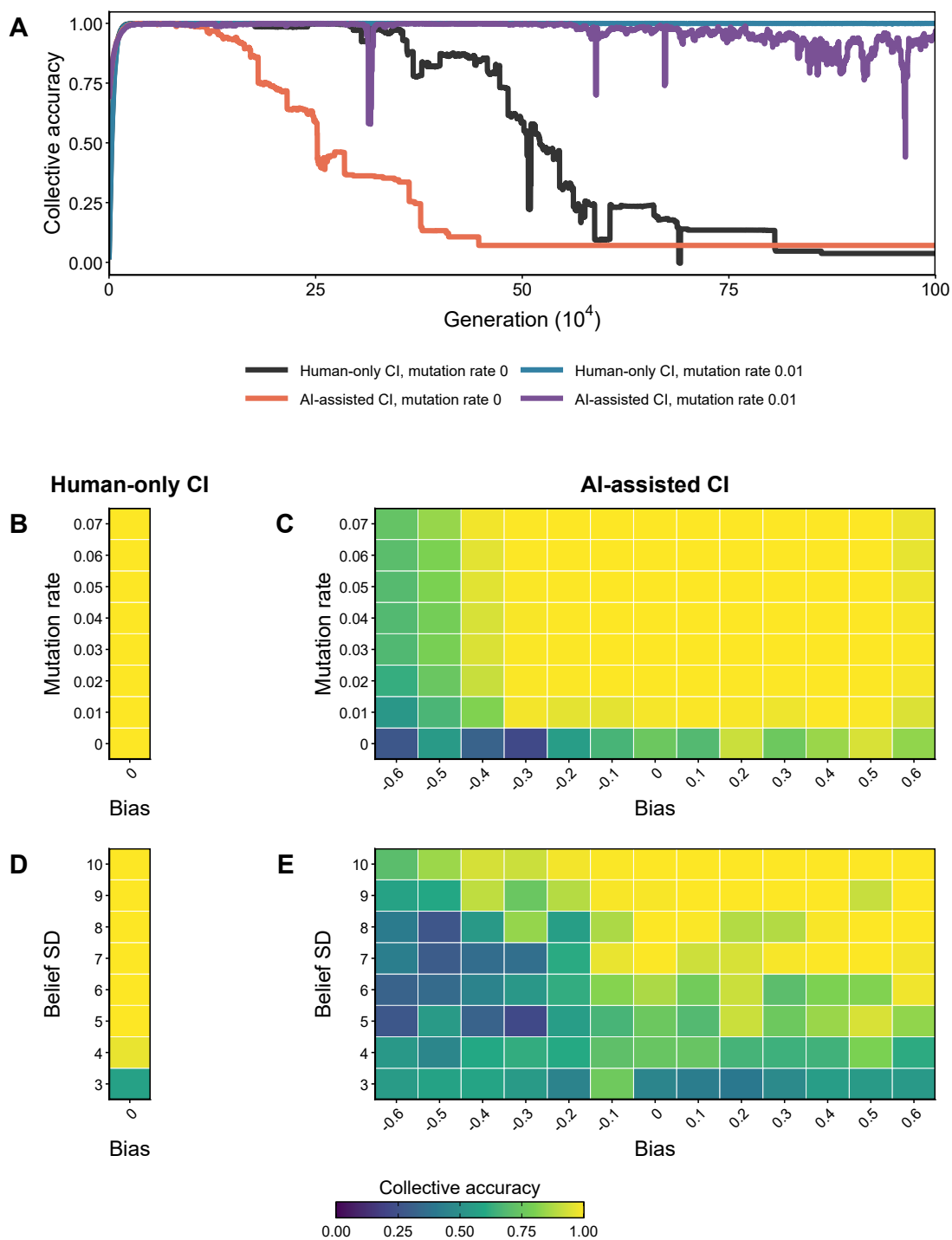


Figure S4: **Caption.** Text.

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